Supplementary Material: On Latent Predictive Residuals and Automated Threshold Calibration for Multivariate Time-Series Anomaly Detection

S-1. Proofs of Propositions

Proof of Proposition 1 (Operator Triviality Invariance). (i) At $\mathbf{K} = \mathbf{0}$ the implemented barrier is saturated at its floor: $r_{\text{safe}} = \epsilon$ is constant on a neighborhood of $\mathbf{0}$, so Φ_{energy} contributes zero gradient there, while the prediction term has gradient $\nabla_{\mathbf{K}}\mathcal{L}_{\text{pred}}\big|_{\mathbf{0}} = -\frac{2}{B}\sum_{b=1}^{B} z_{\text{tgt}}^{(b)} z_{\text{ctx}}^{(b)^T} \neq \mathbf{0}$ by the coupling assumption. Hence $\mathbf{K} = \mathbf{0}$ is not a stationary point and cannot be a local minimizer; the floored penalty remains finite, $\mathcal{R}_{\text{spectral}}(\mathbf{0}) \leq \log D + \beta_{\text{norm}}(\epsilon - 1 - \log\epsilon) \approx \log D + 12.8\,\beta_{\text{norm}}$. (ii) For the unfloored barrier, $-\log r \to +\infty$ as $\|\mathbf{K}\|_F \to 0^+$ while $\mathcal{L}_{\text{pred}} \to E_{\text{tgt}} < \infty$, so $\mathcal{L}(\mathbf{K}) \to +\infty$; the identity operator satisfies $\Delta\mathcal{H}(\mathbf{I}) = \Phi_{\text{energy}}(\mathbf{I}) = 0$ with strictly finite loss, so $\mathbf{K} = \mathbf{0}$ cannot be a global minimizer in the idealization either. The divergence statement (ii) applies to the unfloored limit only; the anti-triviality guarantee for the trained objective rests on (i). □

Proof of Proposition 2. Suppose $\gamma(t)$ is a periodic orbit with period $T_p > 0$, i.e. $\gamma(t + T_p) = \gamma(t)$. Integrating the time derivative of Φ along the closed path yields $\int_0^{T_p} \dot{\Phi}(\gamma(t))\,dt = 0$, since $\Phi(\gamma(T_p)) = \Phi(\gamma(0))$. By the chain rule, $\dot{\Phi}(\gamma(t)) = \langle \nabla\Phi, \dot{\gamma}\rangle$, which equals $-\|\nabla\Phi(\gamma(t))\|_2^2 \leq 0$. The integral of a non-positive integrand vanishes only if $\|\nabla\Phi(\gamma(t))\|_2 \equiv 0$ on $[0, T_p]$, hence $\dot{\gamma}(t) \equiv \mathbf{0}$: γ is an isolated fixed point, contradicting a non-constant periodic orbit. □

S-2. Evaluation Protocol Matrix

A. Streaming Scoring and Calibration Pseudocode

B. Streaming Scoring and Calibration Pseudocode

C. Controlled Same-Backbone Comparison Table

S-3. Auxiliary Input-Channel Saliency Gate: Specification and Studies

A. Formal Specification

To prevent dimensional drowning in input space without target-window leakage, the strictly causal *Causal Channel Saliency Gate* computes channel dispersion strictly from historical context $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$:

$$v_k^{\text{ctx}} = \text{Var}_{\tau \in [t-C, t]}(x_{\tau,k}) = \frac{1}{C-1}\sum_{\tau=t-C}^{t}\left(x_{\tau,k} - \bar{x}_k^{\text{ctx}}\right)^2, \tag{S1}$$

$$s_k = \frac{v_k^{\text{ctx}}}{\sum_{j=1}^{K} v_j^{\text{ctx}} + \epsilon}, \quad g_k = \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right), \tag{S2}$$

where $\sigma(\cdot)$ is the logistic sigmoid, $\epsilon = 10^{-6}$, $\bar{s} = 1/K$ is the uniform variance-share expectation, and $\sigma_s = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(s_k - \bar{s})^2 + \epsilon}$ is the cross-channel dispersion; $(s_k - 1/K)/\sigma_s$ is the spatial z-score of channel energy. The modulation vector $\mathbf{g} \in (0, 1)^K$ scales input channels symmetrically in both encoders, $\tilde{\mathbf{x}}[:, k] = \mathbf{x}[:, k] \cdot g_k$. Because $\mathbf{g}$ depends exclusively on $\mathbf{x}_{\text{ctx}}$, $\partial g_k / \partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ for all k.

Algorithm 1 Streaming Anomaly Scoring and Decision Calibration

Require: Streaming telemetry $\mathbf{X} \in \mathbb{R}^{T \times K}$, filtration $\mathcal{F}_t = \sigma(\mathbf{X}[1 : t, :])$, context horizon $C = 256$, suspect horizon $S = 64$, deployment mode $\mathcal{M} \in \{\text{Causal}, \text{Buffered}\}$, calibrator $\mathcal{C}_{\text{EVT}}$ fit to nominal residuals.

Ensure: Point-level alarm stream $\hat{y}_t \in \{0, 1\}$ for $t \in [C, T]$.

Initialize score buffer $\mathbf{s} \in \mathbb{R}^T \leftarrow \mathbf{0}$, hit counts $\mathbf{c} \in \mathbb{R}^T \leftarrow \mathbf{0}$.
for $t = C$ **to** $T - S$ **do**
  Extract historical context: $\mathbf{x}_{\text{ctx}} \leftarrow \mathbf{X}[t - C : t, :]$ ▷ $\mathcal{F}_t$-measurable
  Encode latent context: $z_{\text{ctx}} \leftarrow E_\theta(\mathbf{x}_{\text{ctx}})$
  Forecast future latent representation: $\hat{z}_{\text{tgt}} \leftarrow P_\psi(z_{\text{ctx}})$
  if $\mathcal{M} =$ Causal (0**-Lookahead Delay) then**
    Target is trailing window: $\mathbf{x}_{\text{tgt}}^{\text{causal}} \leftarrow \mathbf{X}[t - S : t, :]$ ▷ Strictly $\mathcal{F}_t$-measurable
    Encode target: $z_{\text{tgt}} \leftarrow E_{\bar{\theta}}(\mathbf{x}_{\text{tgt}}^{\text{causal}})$
    Compute trailing predictive discrepancy: $s_t \leftarrow \mathcal{D}_{\text{sal}}(\hat{z}_{\text{tgt}} - z_{\text{tgt}})$
    $\hat{y}_t \leftarrow \mathbb{I}(s_t > \tau_{\text{eff}})$ ▷ Instantaneous decision at timestamp t
  else if $\mathcal{M} =$ Buffered (S**-Lookahead Delay) then**
    Extract prospective suspect horizon: $\mathbf{x}_{\text{tgt}} \leftarrow \mathbf{X}[t : t + S, :]$ ▷ $\mathcal{F}_{t+S}$-measurable
    Encode target representation: $z_{\text{tgt}} \leftarrow E_{\bar{\theta}}(\mathbf{x}_{\text{tgt}})$
    Compute forward predictive discrepancy: $\mathcal{D}_t \leftarrow \mathcal{D}_{\text{sal}}(\hat{z}_{\text{tgt}} - z_{\text{tgt}})$
    for $\tau = t$ **to** $t + S - 1$ **do**
      Accumulate suspect overlap: $s_\tau \leftarrow s_\tau + \mathcal{D}_t$, $c_\tau \leftarrow c_\tau + 1$
    end for
    Finalize running average score at step t: $\bar{s}_t \leftarrow s_t / \max(c_t, 1)$
    $\hat{y}_t \leftarrow \mathbb{I}(\bar{s}_t > \tau_{\text{eff}})$ ▷ Finalized at timestamp $t + S$
  end if
end for
return Binary alarm sequence $\hat{\mathbf{y}}$

TABLE S1
MASTER EXPERIMENT MATRIX. DETAILED BREAKDOWN OF ALL EVALUATED STREAM SUBSETS, PARTITION BOUNDARIES, TEMPORAL ISOLATION GAPS, RANDOM SEEDS, AND OPERATIONAL EVALUATION REGIMES ACROSS THE EMPIRICAL INVESTIGATION.

Experimental Suite	Streams	Domains	Partition Split	Isolation Gap	Seeds	Inference Protocol	Primary Hypotheses & Tested Endpoints
Primary Benchmark	33	7	65% / 17.5% / 17.5%	64 steps	5/3†	Buffered ($S = 64$)	Out-of-sample held-out FPR, Point-F1, PR-AUC parity, event recall, and detection delay Δt (Table **??**).
Active Non-GHL Suite	19	6	65% / 17.5% / 17.5%	64 steps	5	Buffered ($S = 64$)	Paired bootstrap 95% CIs and Wilcoxon signed-rank tests on active anomaly dynamics (Table **??**, Table **??**(a)).
Matched Backbone	15	10	80% / 20% nominal	64 steps	1	Buffered ($S = 64$)	Isolates observation recon vs. observation pred vs. latent pred under identical `HybridTCNEncoder` and noise injection (Table S2).
Regularizer Controls	19	6	65% / 17.5% / 17.5%	64 steps	3	Buffered ($S = 64$)	Controlled null hypothesis testing: $\lambda_{\text{ent}} = 0$, $\alpha_{\text{stress}} = 0$, and single-regime controls on fixed encoders (Table **??**(a)).
Strictly Causal Suite	33	7	65% / 17.5% / 17.5%	64 steps	1	Causal (0-lookahead)	Zero-lookahead streaming latency, false alarms per 1k steps (FAR_{1k}), and early event recall (Table **??**, Table S9).
EVT Multi-Parameter	33	7	65% / 17.5% / 17.5%	64 steps	1	Buffered ($S = 64$)	Sensitivity across tail percentiles u, design risks q, kurtosis multipliers β, and non-parametric baselines (Table **??**, Table S11).
Operational Incidents	3	3	Continuous test	None	1	Standard / Sharp / Adaptive	Incident-level evaluation: Point-F1 vs. Event-F1 vs. Range-F1, and dynamic rolling SPOT drift tracking (Table S8).

† Five seeds on the 19 non-GHL streams and GHL training loops 17–18; three seeds on the remaining twelve large GHL streams ($> 10^6$ observations each) owing to training cost.

TABLE S2
CONTROLLED SAME-BACKBONE COMPARISON ISOLATING PREDICTION SPACE UNDER AN IDENTICAL `HYBRIDTCNENCODER` (C=256, S=64, d=32): DATASET-MACRO MEANS ACROSS 15 STREAMS SPANNING TEN DOMAINS, UNDER CLEAN TELEMETRY AND JITTER INJECTION (SNR $\in \{\infty, 20, 10, 0$ DB$\}$).

Model Backbone	Noise SNR	Val γ_2	Test γ_2	Empirical FPR	Mean False Pos.	Point-F1	Recall
TCN-Obs-Recon	Clean (∞)	6.09	57.26	27.29%	10,554.7	0.1977	0.4330
	20 dB	6.09	57.23	27.47%	10,587.1	0.1975	0.4337
	10 dB	6.09	57.14	32.08%	11,168.1	0.1904	0.4546
	0 dB	6.09	57.09	54.10%	17,318.2	0.2184	0.6400
TCN-Obs-Pred	Clean (∞)	6.00	55.66	27.57%	10,455.2	0.1916	0.4315
	20 dB	6.00	56.09	27.93%	10,505.6	0.1915	0.4328
	10 dB	6.00	56.00	32.46%	11,025.9	0.1984	0.4619
	0 dB	6.00	55.66	50.98%	15,558.5	0.2188	0.6414
TS-JEPA (Ours)	Clean (∞)	**0.88**	**1.77**	**11.80%**	**3,949.5** ($\downarrow 62.6\%$)	0.0785	0.1460
	20 dB	**0.88**	**1.67**	**12.25%**	**4,061.3**	0.0741	0.1407
	10 dB	**0.88**	**1.54**	**12.72%**	**4,248.9**	0.0691	0.1300
	0 dB	**0.88**	**1.53**	**13.93%**	**4,269.0**	0.0670	0.1551

B. Gating-Regime Comparison and Fault-Type Sensitivity

Four gating regimes were trained under an identical TS-JEPA configuration (C=256, S=64, seed 42, SPOT q=10^{-3} on held-out nominal telemetry): *context-derived gating* ($\mathbf{g} = g(\mathbf{x}_{\text{ctx}})$ applied symmetrically; strictly causal), a non-causal *target-only leaky control* ($\mathbf{g} = g(\mathbf{x}_{\text{tgt}})$, upper-bounding the benefit of violating causality), *independent* context/target gating (asymmetric; breaks latent metric alignment), and *uniform pooling* (deployed). The effect is stream-dependent in sign and magnitude: the context gate substantially improves MSL ($0.047 \rightarrow 0.227$), where anomalies concentrate in few coordinates, but degrades SMAP ($0.280 \rightarrow 0.019$), where fault signatures are distributed. Macro Point-F1 ranks uniform (0.118) above context gating (0.077), with the leaky control at 0.128 and independent gating weakest (0.071). Since the leaky control adds only $+8.6\%$ relative over causal gating, most of the effect reflects quiescent-noise suppression rather than target statistics.

For per-fault-type behavior, six single-channel fault classes (flatline, stuck sensor, dropout, low-amplitude step, spike control, mid-variance flatline) were injected into six multivariate streams under four gating modes. Two findings emerge. First, the context-derived variance gate approximately doubles point recall relative to uniform pooling (macro 0.045 vs. 0.023) at roughly three times the nominal false-positive rate (0.069 vs. 0.019); a robust-MAD variant is marginally stronger (0.060 recall). Second, the increase concentrates on transient signatures, while quiet faults (flatline, stuck sensor, dropout) remain weakly detected under *every* gating regime (point recall 0.02–0.07). The blind spot therefore reflects the aggregate

latent discrepancy signal on wide telemetry rather than a deficiency of gating alone.

TABLE S3
AUXILIARY INPUT-CHANNEL GATING VARIANT (NOT DEPLOYED): POINT-F1 UNDER UNIFORM, CONTEXT-GATE, TARGET-ONLY-LEAKY, AND INDEPENDENT GATING.

Stream	Uniform (deployed)	Context Gate	Target-Only	Independent
NASA MSL `M-1` ($K = 55$)	0.047	**0.227**	0.051	0.070
NASA SMAP `P-3` ($K = 25$)	**0.280**	0.019	0.155	0.055
GHL Loop `17` ($K = 23$)	0.000	0.000	0.000	0.000
Daphnet `S01R01E1` ($K = 9$)	0.146	0.061	**0.306**	0.158
Macro Avg	**0.118**	0.077	0.128	0.071

TABLE S4
MACRO POINT RECALL OF GATING MODES UNDER INJECTED SINGLE-CHANNEL FAULTS.

Injected Fault Type	Uniform	Var. Gate	Robust-MAD
Flatline	0.021	0.043	**0.058**
Flatline (mid-var ch.)	0.024	0.048	**0.068**
Stuck sensor	0.021	0.043	**0.058**
Dropout	0.020	0.043	**0.058**
Low-amplitude step	0.030	0.049	**0.062**
$+5\sigma$ spike control	0.023	0.042	**0.059**
Mean nominal FPR	**0.019**	0.069	0.055

S-4. MATHEMATICAL DIAGNOSTICS OF PHYSICAL REGULARIZERS

To understand why the proposed regularizers enforce their intended mathematical properties yet fail to improve downstream anomaly detection, we conduct targeted dynamical diagnostic experiments.

1) Operator-Entropy: Anti-Collapse Spectral Rank Preservation: In Operator-Entropy JEPA, the transition operator $K \in \mathbb{R}^{D\times D}$ ($D = 32$) is regularized via Von Neumann spectral entropy $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ on normalized Gram matrix $\rho = KK^T/\mathrm{Tr}(KK^T)$. Table S5 evaluates effective operator rank via the RankMe metric ($\exp(\mathcal{H})$ where $\mathcal{H}$ is the spectral entropy of singular values), representation covariance conditioning, and downstream calibrated Point-F1 across five telemetry streams.

As shown in Table S5, unconstrained Vanilla TS-JEPA and VICReg collapse across 7–8 singular modes, yielding an effective operator rank of 24.3–24.8. Operator-Entropy JEPA achieves an effective operator rank of 31.996 out of 32.0, confirming that the regularizer prevents modal collapse. However, full operator rank does not improve downstream anomaly detection: telemetry anomalies are frequently localized subsystem deviations occurring in low-dimensional subspaces. Preserving high-rank transition dynamics forces the model to track ambient high-frequency variations, which does not enhance fault discrimination.

2) Reynolds-Stress: Covariance Alignment vs. Marginal Perturbations: Reynolds-Stress JEPA aligns second-order latent fluctuation statistics ($\Sigma_{z'} = \frac{1}{B}\sum z'(z')^T$) between predicted and target horizons. Table S6 compares models on synthetic telemetry isolating pure cross-channel correlation flips ($\rho : +0.85 \rightarrow -0.85$) from pure marginal variance scaling ($3.0\times$).

TABLE S5
EMPIRICAL SPECTRAL DIAGNOSTICS OF LEARNED TRANSITION OPERATORS AND REPRESENTATIONS ACROSS FIVE BENCHMARK TELEMETRY STREAMS ($D = 32$). OPERATOR-ENTROPY PRESERVES FULL THEORETICAL RANK ($\exp(\mathcal{H}) = 31.996$ OUT OF 32.0), PREVENTING DIMENSIONAL COLLAPSE.

Model Architecture	Operator Rank (RankMe)	Rep. Rank	Point-F1	Empirical FPR
Operator-Entropy JEPA	**31.996**	3.564	**0.0710**	0.1497
Representation-Entropy JEPA	24.647	**7.579**	0.0479	0.0442
Vanilla TS-JEPA (Unconstrained)	24.807	3.993	0.0301	**0.0371**
VICReg TS-JEPA	24.339	5.881	0.0043	0.0273

TABLE S6
CONTROLLED COVARIANCE VS. MARGINAL ANOMALY BENCHMARK. REYNOLDS-STRESS JEPA IMPROVES MARGINAL VARIANCE DETECTION OVER UNCONSTRAINED TS-JEPA (0.5838 VS. 0.3313 PR-AUC), BUT BOTH LATENT MODELS STRUGGLE ON PURE CORRELATION FLIPS.

	Pure Covariance Shift		Pure Marginal Shift	
Model Architecture	**PR-AUC**	**Point-F1**	**PR-AUC**	**Point-F1**
Reynolds-Stress JEPA	0.0498	0.0000	0.5838	**0.0593**
TS-JEPA (Unconstrained)	0.0409	0.0000	0.3313	0.0000
TimesNet	**0.9313**	0.6558	**0.9998**	0.6267
TranAD	0.6761	**0.7246**	0.9343	**0.6560**

Table S6 indicates that on marginal variance shifts, Reynolds-Stress improves PR-AUC by $+76\%$ over TS-JEPA (0.5838 vs. 0.3313). However, on pure covariance shifts without marginal changes, both latent models show low sensitivity (0.041–0.050 PR-AUC) relative to observation-space baselines (0.676–0.931). Standard temporal pooling across channels averages out instantaneous phase alignments, preventing the second-moment loss from isolating cross-channel relationship shifts unless explicit pairwise relational graphs are modeled.

3) Potential-Flow: Conservative vs. Circulating Dynamics: Potential-Flow JEPA constrains transition velocities to curl-free potential gradients ($\hat{v} = -\nabla\Phi$). Table S7 evaluates Potential-Flow against Helmholtz-JEPA (decomposing velocity into gradient plus solenoidal circulation: $\hat{v} = -\nabla\Phi + \mathrm{curl}(A)$) and unconstrained TS-JEPA across two dynamical benchmarks: a conservative Duffing oscillator and a non-conservative Van der Pol limit-cycle attractor.

Table S7 confirms that pure curl-free potential flow is mathematically incapable of representing closed circulating orbits, yielding poor detection on Van der Pol (0.3548 Point-F1 vs. 0.7574 for Helmholtz JEPA). Because physical telemetry often exhibits non-conservative, cyclic behavior (e.g., diurnal cycles, rotational turbines), enforcing irrotational dynamics severely penalizes normal circulating trajectories, explaining Potential-Flow's poor benchmark performance (0.065 dataset-macro F1 in Table **??**).

4) Multiscale Front-End: Dyadic-Shell Construction: The encoder front-end constructs $J{=}4$ dyadic shells on an $N_{\text{fft}} = 256$ real-FFT grid ($F = 129$ non-negative bins $\omega_m = m/N_{\text{fft}}$). Dyadic center frequencies sit at $c_j = 2^{j-J+1}$, $j \in \{0, 1, 2, 3\}$. Unnormalized profiles combine a half-Gaussian anchored at DC ($j = 0$), log-frequency octave bells $\tilde{\Psi}_j(\omega_m) = \exp(-(\log_2 \omega_m - \log_2 c_j)^2/2\sigma^2)$ with bandwidth $\sigma = 0.6$ ($j \in \{1, 2\}$), and a plateau toward Nyquist ($j = J - 1$),

TABLE S7
DYNAMICAL SYSTEM VALIDATION COMPARING CURL-FREE POTENTIAL-FLOW ($\hat{v} = -\nabla\Phi$), HELMHOLTZ DECOMPOSITION ($\hat{v} = -\nabla\Phi + \text{CURL}(A)$), AND UNCONSTRAINED TS-JEPA ON CONSERVATIVE HAMILTONIAN VS. NON-CONSERVATIVE LIMIT-CYCLE PHASE SPACES.

	Conservative Duffing		Non-Conservative Van der Pol	
Model Architecture	**PR-AUC**	**Point-F1**	**PR-AUC**	**Point-F1**
Helmholtz JEPA ($\nabla\Phi + \text{curl}(A)$)	**0.5985**	**0.7815**	**0.5250**	**0.7574**
TS-JEPA (Unconstrained)	0.5855	0.7583	0.4577	0.6660
Potential-Flow JEPA ($-\nabla\Phi$)	0.1701	0.2036	0.3106	0.3548

normalized bin-wise into an exact partition of unity:

$$\Psi_j[\omega_m] = \frac{\tilde{\Psi}_j(\omega_m)}{\sum_{l=0}^{J-1} \tilde{\Psi}_l(\omega_m)}, \tag{S3}$$
$$\sum_{j=0}^{J-1} \Psi_j[\omega_m] \equiv 1.0.$$

The split is applied as $\mathbf{x}_j = \text{irfft}(\Psi_j \odot \text{rfft}(\mathbf{x}))$ (PoU deviation $< 1.2 \times 10^{-7}$, reconstruction error $< 5 \times 10^{-8}$). Each shell feeds a full (non-depthwise) causal 1D convolution with kernels $k \in \{15, 9, 5, 3\}$ and causal padding $p = k - 1$, recombined by a softmax router over pooled per-shell energies, a 1×1 projection, residual scaling $\gamma = 0.1$, and LayerNorm. Because shells overlap smoothly, band energies are correlated ($\sum_j \|\mathbf{x}_j\|^2$ recovers $\approx 70\%$ of signal energy) and the finest shell concentrates near Nyquist; per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (`reports/lp_filterbank_diagnostics.csv`).

S-5. OPERATIONAL CASE STUDIES AND COMPUTATIONAL PROFILE

A. Operational Incident Evaluation: Event Metrics and Adaptation

Evaluating telemetry solely by point-level metrics can penalize slight temporal boundary offsets in continuous anomalies. In operational systems, faults represent extended continuous incidents; a detector that identifies the event with a minor phase lag is penalized on every out-of-boundary point.

To analyze operational incident detection, Table S8 evaluates three complementary metrics across three telemetry streams: Room Occupancy (stream: `default`, 11 events), NASA SMAP (stream: `P-3`, 1 event), and CalIt2 (stream: `traffic`, 21 events). We compare strict Point-F1, Event-F1 ($P_{\text{event}}, R_{\text{event}}$), and Tatbul et al. Range-Aware F1 ($F1_{\text{range}}$) across three operational inference configurations: Standard (middle-window, static EVT), Sharp (dense causal, static EVT), and Adaptive (dense, dynamic rolling SPOT).

Table S8 highlights three operational characteristics: First, static EVT thresholding achieves 100% event recall on Room Occupancy and SMAP (`P-3`), but generates over 4,000 false alarm points on Room Occupancy due to unmodeled diurnal baseline shifts, causing point precision to collapse to 20.0%. Second, dynamic rolling SPOT suppresses false alarms by $21\times$ (from 4,705 to 223 points) and raises precision to 29.9%, but event recall drops from 100% to 63.6%. Tracking non-stationary drift via rolling quantiles creates an operational trade-off between alarm fatigue and missed events. Third, on transient burst streams (CalIt2, 21 short spikes), models detect only 4.8%–23.8% of events under rolling SPOT, demonstrating that short pulses remain difficult to distinguish from ambient noise.

B. Five-Stream Causal-vs-Buffered Operational Comparison

Table S9 compares operational latency, false-alarm rates, and detection delay across five matched telemetry streams (NASA SMAP `P-3`, SMD `machine-1-2`, NASA MSL `M-1`, Daphnet `S01R01E1`, and GECCO `water_quality`) under the strictly causal (0-lookahead) and buffered (64-lookahead) regimes defined in Section **??**.

Under strictly causal streaming, TS-JEPA maintains an empirical false-positive rate of 4.18% (41.76 false alarms per 1,000 points), generating an average of 1,685.6 false alarms per stream. In contrast, TimesNet and TranAD generate 16,952.0 and 16,637.6 false alarms (empirical FPRs of 30.06% and 20.81%). On GECCO water telemetry (113,000 test points), TimesNet and TranAD trigger over 81,500 false alarms (99.6% FPR), whereas TS-JEPA triggers zero false alarms.

Two methodological trade-offs govern these results: First, the false-alarm reduction under strict causality comes at the expense of early event capture. TS-JEPA detects 10.7% of anomaly events within $\tau \leq 64$ steps under causal streaming, compared to 22.9% for TimesNet and 32.9% for TranAD.

C. Computational Profile: Complexity, Latency, and Memory Footprint

Table S10 benchmarks parameter complexity, peak GPU memory footprint (VRAM), forward inference latency, and streaming throughput across all evaluated architectures on an NVIDIA RTX 5090 GPU (input dimension $K = 25$, batch size 32, context $C = 256$, suspect $S = 64$).

As shown in Table S10, Operator-Entropy JEPA possesses fewer trainable parameters (285,733) than unconstrained TS-JEPA (289,061), because its explicit Koopman transition matrix $K \in \mathbb{R}^{32\times32}$ (1,024 weights) replaces the multi-layer MLP predictor. All JEPA variants maintain a compact peak GPU memory footprint of 183.8–185.3 MB, consuming nearly $7\times$ less memory than TranAD (1,280.0 MB), whose self-attention maps scale quadratically with window length. Forward inference latency for the JEPA family ranges between 0.60–0.84 ms per window (1,184–1,647 windows per second). While TimesNet (0.29 ms) and TranAD (0.19 ms) achieve faster forward inference, the JEPA family operates well within sub-millisecond operational streaming deadlines on GPU hardware. These measurements characterize workstation accelerator throughput; on resource-constrained embedded processors or flight microcontrollers lacking dedicated tensor cores, sequential end-to-end latency depends directly on clock frequencies and memory bandwidth, motivating integer quantization (Section **??**) for edge deployment.

TABLE S8

OPERATIONAL INCIDENT DETECTION COMPARISON ACROSS REPRESENTATIVE TELEMETRY STREAMS: ROOM OCCUPANCY (STREAM: `default`, 11 GROUND-TRUTH EVENTS), NASA SMAP (STREAM: `P-3`, 1 THRUSTER FAULT), AND CALIT2 (STREAM: `traffic`, 21 PEDESTRIAN BURST EVENTS). WE CONTRAST STANDARD (MIDDLE, STATIC EVT), SHARP (DENSE, STATIC EVT), AND ADAPTIVE (DENSE, DYNAMIC SPOT) INFERENCE MODES.

Evaluated Stream	Operational Configuration	Pt-F1	Pt-Prec	Evt-F1	Evt-Prec	Evt-Rec	Rng-F1	False Pos.
Room Occupancy: Stream `default` (Environmental Sensing, 11 Ground-Truth Incidents)								
`operator_entropy`	Standard (Middle, Static EVT)	0.3728	22.9%	0.5238	35.5%	100%	0.3708	3,842
`operator_entropy`	Sharp (Dense, Static EVT)	0.3329	20.0%	**0.9167**	**84.6%**	**100%**	0.3329	4,705
`operator_entropy`	Adaptive (Dense, Dynamic SPOT)	0.1273	29.9%	0.6364	63.6%	63.6%	0.3699	**223** ($\downarrow 21\times$)
`potential_flow`	Sharp (Dense, Static EVT)	0.3431	20.7%	**0.9167**	**84.6%**	**100%**	0.3431	4,496
`potential_flow`	Adaptive (Dense, Dynamic SPOT)	0.0441	10.2%	0.5714	60.0%	54.5%	0.1636	**291** ($\downarrow 15\times$)
NASA SMAP: Stream `P-3` (Spacecraft Telemetry, 1 Continuous Fault Segment)								
`operator_entropy`	Standard (Middle, Static EVT)	0.0087	14.6%	0.1818	10.0%	100%	0.2266	35
`operator_entropy`	Sharp (Dense, Static EVT)	0.1274	27.3%	0.1538	8.3%	100%	**0.3634**	295
`potential_flow`	Adaptive (Dense, Dynamic SPOT)	0.0110	6.9%	**0.3333**	20.0%	**100%**	0.1213	**108**
CalIt2: Stream `traffic` (Pedestrian Entrance Flow, 21 Transient Event Clusters)								
`operator_entropy`	Standard (Middle, Static EVT)	0.0131	1.9%	0.0588	7.7%	4.8%	0.0216	53
`operator_entropy`	Adaptive (Dense, Dynamic SPOT)	**0.1368**	8.7%	**0.3226**	50.0%	23.8%	**0.1274**	357
`potential_flow`	Adaptive (Dense, Dynamic SPOT)	0.0903	14.3%	0.0909	**100.0%**	4.8%	0.0714	**42**

TABLE S9

OPERATIONAL LATENCY AND DETECTION FIDELITY UNDER STRICTLY CAUSAL (0-LOOKAHEAD DELAY) VERSUS BUFFERED WINDOW (64-LOOKAHEAD DELAY) INFERENCE ACROSS 5 REPRESENTATIVE TELEMETRY STREAMS. UNDER CAUSAL STREAMING, TS-JEPA ACHIEVES AN EMPIRICAL FALSE-POSITIVE RATE OF 4.18% ($\text{FAR}_{1\text{K}} = 41.76$ FALSE ALARMS PER 1K POINTS), COMPARED TO 30.06% FOR TIMESNET AND 20.81% FOR TRANAD.

Inference Setting	Model	Empirical FPR	$\text{FAR}_{1\text{k}}$ (/1k pts)	False Positives	Point-F1	Precision	Mean Delay (Δt)	Recall ($\tau \leq 64$)
Strictly Causal (0-Lookahead)	TS-JEPA	**0.0418**	**41.76**	**1,685.6**	0.0688	**0.1581**	130.3	0.1071
	TimesNet	0.3006	300.63	16,952.0	0.0774	0.1525	251.2	0.2286
	TranAD	0.2081	208.13	16,637.6	**0.0881**	0.1255	**43.2**	**0.3286**
Buffered Window (64-Lookahead)	TS-JEPA	**0.1514**	**151.39**	**2,696.0**	0.0431	0.0355	89.6	0.1357
	TimesNet	0.2800	280.03	16,916.8	**0.0807**	0.1404	235.3	0.2571
	TranAD	0.1998	199.82	16,339.6	0.0197	**0.2284**	**32.8**	**0.2786**

TABLE S10

COMPUTATIONAL PROFILE, MEMORY CONSUMPTION, LATENCY, AND THROUGHPUT BENCHMARKED ON AN NVIDIA RTX 5090 GPU ($K = 25$, WINDOW SIZE 320).

Model Architecture	Parameters	VRAM (MB)	Latency (ms/win)	Infer (win/s)	Train (win/s)
TS-JEPA	289,061	**183.8**	0.6072	1,646.8	612.8
Operator-Entropy JEPA	**285,733**	183.9	0.6939	1,441.0	553.2
Reynolds-Stress JEPA	295,953	184.0	0.7456	1,341.2	535.8
Potential-Flow JEPA	338,117	184.4	0.8445	1,184.2	400.2
Helmholtz JEPA	416,391	185.3	0.8378	1,193.6	526.6
TimesNet	648,553	279.6	**0.2908**	3,439.4	1,161.4
TranAD	205,746	1,280.0	**0.1897**	**5,272.2**	**1,554.4**

S-6. EXTENDED EVT SENSITIVITY AND FAILURE-MECHANISM DECOMPOSITION

Table S11 provides a multi-parameter sensitivity analysis of the extreme-value calibration procedure across varying initial tail thresholds u, theoretical design risk levels q, and kurtosis-adaptive coefficients β. Three key findings emerge: First, varying the initial threshold from $u = 0.95$ to $u = 0.99$ produces negligible changes in out-of-sample test FPR (17.27%–17.36%) and Point-F1 (0.1028–0.1030), indicating that GPD tail estimates are stable with respect to initial tail percentile selection. Second, scaling the theoretical design risk q by two orders of magnitude (from 10^{-2} to 10^{-4}) shifts validation FPR from 0.0074 to 0.00009, but test FPR shifts by less than 1.7% (from 18.95% to 17.27%), demonstrating that operational distribution drift completely dominates the theoretical design parameter. Third, increasing the kurtosis adjustment coefficient β systematically degrades performance: at $\beta = 0.083$, held-out validation FPR jumps from 0.0019 to

TABLE S11

SENSITIVITY ANALYSIS OF EXTREME VALUE THEORY (SPOT) CALIBRATION PARAMETERS ACROSS BENCHMARK STREAMS. WE EVALUATE THE SENSITIVITY OF EMPIRICAL FALSE POSITIVE RATES (ON HELD-OUT NOMINAL VALIDATION SPLITS AND TEST SEQUENCES) AND POINT-F1 ACROSS INITIAL TAIL QUANTILES $u \in \{0.95, 0.98, 0.99\}$, DESIGN RISK PARAMETERS $q \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, AND KURTOSIS ADAPTATION COEFFICIENTS $\beta \in \{0.0, 0.083, 0.167, 0.25, 0.50\}$ ALONGSIDE NON-PARAMETRIC BASELINES.

Threshold Calibration Method	Design q	Tail u	Val FPR	Test FPR	Point-F1
Initial Tail Quantile and Risk Level Sensitivity (Static SPOT)					
Static SPOT ($u = 0.95, q = 10^{-4}$)	10^{-4}	0.95	0.00009	0.1727	0.1029
Static SPOT ($u = 0.95, q = 10^{-3}$)	10^{-3}	0.95	0.00190	0.1762	0.1030
Static SPOT ($u = 0.95, q = 10^{-2}$)	10^{-2}	0.95	0.00741	0.1895	0.1020
Static SPOT ($u = 0.98, q = 10^{-4}$)	10^{-4}	0.98	0.00009	0.1733	0.1028
Static SPOT ($u = 0.98, q = 10^{-3}$) (Baseline)	10^{-3}	0.98	0.00190	0.1764	0.1030
Static SPOT ($u = 0.98, q = 10^{-2}$)	10^{-2}	0.98	0.00743	0.1886	0.1022
Static SPOT ($u = 0.99, q = 10^{-4}$)	10^{-4}	0.99	0.00017	0.1736	0.1030
Static SPOT ($u = 0.99, q = 10^{-3}$)	10^{-3}	0.99	0.00210	0.1763	0.1029
Static SPOT ($u = 0.99, q = 10^{-2}$)	10^{-2}	0.99	0.00703	0.1862	0.1029
Kurtosis-Adaptive Multiplier Sensitivity ($\tau_{\text{eff}} = \tau(1 + \beta \tanh \gamma_2)$)					
Kurtosis-Adaptive ($\beta = 0.000$, Standard)	10^{-3}	0.98	0.00190	0.1764	0.1030
Kurtosis-Adaptive ($\beta = 0.083$)	10^{-3}	0.98	0.27780	0.2241	0.0973
Kurtosis-Adaptive ($\beta = 0.167$)	10^{-3}	0.98	0.30463	0.2686	0.0964
Kurtosis-Adaptive ($\beta = 0.250$)	10^{-3}	0.98	0.33991	0.3813	0.0995
Kurtosis-Adaptive ($\beta = 0.500$)	10^{-3}	0.98	0.54032	0.6096	0.2560
Moors Quantile Kurtosis ($\beta = 0.167$)	10^{-3}	0.98	0.19350	0.1848	0.1024
Non-Parametric Validation Baselines					
Max-of-Validation	–	–	**0.00000**	**0.1774**	0.1029
Empirical 99.5th Percentile	–	–	0.00544	0.1837	0.1028
Gaussian 3-Sigma ($\mu + 3\sigma$)	–	–	0.00882	0.1879	0.1016

0.2778, and at $\beta = 0.50$, it collapses to 0.5403 (test FPR 60.96%), flooding the system with alarms. Non-parametric validation rules (such as Max-of-Validation and Empirical 99.5th Percentile) achieve comparable or superior nominal calibration (0.0000–0.0054 validation FPR) without requiring parametric tail assumptions.

To diagnose why extreme-value calibration breaks down on operational telemetry, Table S12 isolates four statistical mechanisms across benchmark telemetry streams:

1) **Residual Autocorrelation and Extremal Clustering:** The Pickands-Balkema-de Haan theorem [1] requires mutually independent tail exceedances. However, reconstructive observation residuals exhibit serial correlation, with lag-1 autocorrelation $\rho_1 = 0.983 \pm 0.014$ for TimesNet and $\rho_1 = 0.990 \pm 0.007$ for TranAD. Using the Ferro-Segers extremal index estimator [2], observation exceedances form persistent temporal clusters with mean cluster sizes of $1/\theta = 12.7$ to 13.3 steps (reaching 45.0 steps on GECCO water quality telemetry). Clustering collapses the effective independent tail sample size ($N_{\text{eff}} = \theta N_u$) from $N_u = 77$–91 observations down to $N_{\text{eff}} \approx 2$–6 independent clusters. In contrast, latent prediction compresses lag-1 autocorrelation to $\rho_1 = 0.820 \pm 0.135$ and mean cluster size to 5.2 ± 5.8 steps.
2) **Non-Stationary Distribution Drift:** Non-stationarity drives a wedge between calibration and held-out evaluation splits. Two-sample Kolmogorov-Smirnov tests reveal substantial drift in reconstructive models ($D_{\text{KS}} = 0.357 \pm 0.178$ for TimesNet; $D_{\text{KS}} = 0.388 \pm 0.284$ for TranAD; $p < 10^{-13}$ across streams). This drift shifts the 98th percentile of nominal residuals by $+470.8\%$ on TimesNet and $+48.6\%$ on TranAD between splits. Conversely, TS-JEPA maps temporal sequences into an abstract representation space where environmental shifts are compressed, yielding less than half the KS distance ($D_{\text{KS}} = 0.154 \pm 0.069$) and near-zero relative tail drift ($\Delta q_{98} = -0.36\% \pm 14.1\%$).
3) **GPD Goodness-of-Fit Degradation:** Cramér-von Mises tests for Generalized Pareto tail convergence reject the stationary GPD null hypothesis ($p < 0.05$) across autocorrelated observation streams. Because parameter estimation fits a stationary tail model to non-stationary, autocorrelated bursts, parametric SPOT thresholds provide no advantage over simple empirical validation percentiles.
4) **Empirical Risk Overshoot:** Against the specified design risk $q = 10^{-3}$ (0.1%), TimesNet generates a held-out nominal FPR of $4.32\% \pm 7.60\%$ ($43.2\times$ overshoot), and TranAD produces $4.76\% \pm 10.43\%$ ($47.6\times$ overshoot). By suppressing residual drift and tail clustering, TS-JEPA curbs the empirical nominal FPR to $0.33\% \pm 0.59\%$ ($3.3\times$ overshoot), reducing the discrepancy between theoretical risk and operational deployment by an order of magnitude.

S-7. Benchmark Suite and Stream Detail

S-8. Revision Experiments

This section collects the four analyses added in response to the review: matched-backbone residual diagnostics, detection quality at matched false-positive-rate budgets, a controlled-drift calibration study, and training-free nonparametric references.

TABLE S12

EMPIRICAL DECOMPOSITION OF EXTREME VALUE CALIBRATION FAILURE. WE ISOLATE THE FOUR HYPOTHESIZED DRIVERS OF THRESHOLD BREAKDOWN ACROSS BENCHMARK TELEMETRY STREAMS: (1) RESIDUAL AUTOCORRELATION (ρ_1) AND EXTREMAL CLUSTERING ($\theta, 1/\theta$), WHICH REDUCE THE EFFECTIVE INDEPENDENT TAIL SAMPLE SIZE ($N_{\text{EFF}} = \theta N_u$); (2) NON-STATIONARY DISTRIBUTION DRIFT BETWEEN CALIBRATION AND HELD-OUT EVALUATION SPLITS (KOLMOGOROV-SMIRNOV D_{KS} AND RELATIVE 98TH-PERCENTILE DRIFT $\Delta q_{98}\%$); AND (3) ACTUAL HELD-OUT NOMINAL FALSE-POSITIVE RATE AND EMPIRICAL RISK OVERSHOOT RELATIVE TO THE THEORETICAL DESIGN RISK $q = 10^{-3}$ (0.1%). OBSERVATION-SPACE MODELS EXHIBIT LARGE DISTRIBUTION DRIFT ($\Delta q_{98} = +470\%$) AND SEVERE CLUSTERING ($1/\theta \approx 12.7$–13.3 STEPS), CAUSING A $43\times$ TO $47\times$ OVERSHOOT OF DESIGN RISK. IN CONTRAST, LATENT PREDICTION COMPRESSES DISTRIBUTION DRIFT ($\Delta q_{98} = -0.36\%$), REDUCING RISK OVERSHOOT BY AN ORDER OF MAGNITUDE.

Architecture & Objective	**Autocorr.** (ρ_1) (Lag-1)	**Extremal Index** (θ) (Ferro-Segers)	**Mean Cluster** ($1/\theta$ steps)	**KS Drift** (D_{KS}) (Cal vs. Eval)	**Held-Out FPR** (Nominal Target: 0.1%)	**Risk Overshoot** ($\text{FPR}_{\text{eval}}/10^{-3}$)
TimesNet (Reconstructive)	0.983 ± 0.014	0.528 ± 0.518	12.7 ± 17.1	0.357 ± 0.178	$4.32\% \pm 7.60\%$	$43.2 \times \pm 76.0\times$
TranAD (Reconstructive)	0.990 ± 0.007	0.484 ± 0.483	13.3 ± 17.1	0.388 ± 0.284	$4.76\% \pm 10.43\%$	$47.6 \times \pm 104.3\times$
TS-JEPA (Latent Predictive)	0.820 ± 0.135	0.483 ± 0.356	5.2 ± 5.8	0.154 ± 0.069	$0.33\% \pm 0.59\%$	$3.3 \times \pm 5.9\times$

TABLE S13

BENCHMARK SUITE SPECIFICATIONS: 45 MATCHED TELEMETRY STREAMS ACROSS 11 DATASETS (>5.06M OBSERVATIONS).

Dataset	**Physical Sensing Domain**	**Streams (Held / Tot)**	**Dim** K	N_{train}	N_{test}	**Anom % (π)**	**Primary Anomaly Typologies**	**Data Partition Protocol**
Domain 1: Satellite and Spacecraft Remote Telemetry								
NASA SMAP [3]	Soil Moisture Satellite Subsystems	6 / 6	25	17,022	50,834	15.18%	Subsystem power drops, thermal drifts, cmd faults	65/17.5/17.5 held-out nominal split
NASA MSL [3]	Curiosity Rover Robotic Actuators	4 / 6	55	12,331	13,566	22.88%	Actuator stalls, motor spikes, sensor freezes	65/17.5/17.5 held-out (2 short excluded)
SWAN-SF [4]	Solar Photospheric Magnetograms	1 / 1	38	19,672	72,000	27.92%	Active region magnetic field eruptive bursts	65/17.5/17.5 held-out nominal split
Domain 2: Wearable Biometrics and Kinematics								
Daphnet FoG [5]	Wearable Body Sensor Accelerometry	6 / 6	9	48,240	217,344	13.34%	Freezing-of-Gait motor breakdown, limb tremors	65/17.5/17.5 held-out nominal split
Domain 3: Industrial Processes and Cyber-Physical Systems								
GHL Heating Loop	Thermal and Fluid Transport Loop	14 / 14	16	1,645,137	1,154,014	1.35%	Valve sticking, heater failures, pressure surge	65/17.5/17.5 held-out nominal split
Genesis Automation	Robotic Pick-and-Place Demonstrator	1 / 1	18	3,244	9,332	0.30%	Mechanical manipulation jams, gripper slip	65/17.5/17.5 held-out nominal split
GECCO Water IoT	Hydraulic and Water Distribution	0 / 1	9	27,560	83,113	1.54%	Chemical contamination, line bursts, pH drops	Operational case study ($> 10^6$ steps)
Domain 4: Smart Infrastructure and Environmental Sensing								
Room-Occupancy [6]	Ambient Environmental Sensing	0 / 2	5	2,565	14,317	19.31%	Room-occupancy transitions, airflow anomalies	Incident case study (nominal too short)
CICIDS Edge Network	Edge Gateway Flow Telemetry	0 / 6	72	278,722	1,343,920	23.91%	DDoS volume surges, brute force, port scans	Extended archive ($> 10^6$ steps)
CalIt2 [7]	Optical People-Flow Counter	0 / 1	2	940	4,032	3.42%	Crowd attendance surges, building ingress bursts	Incident case study (nominal too short)
Metro Traffic	Highway Loop Detector Volume	1 / 1	5	24,072	24,102	0.13%	Highway congestion, accidents, severe slowdowns	65/17.5/17.5 held-out nominal split
Total / Benchmark Suite	**4 Diverse Physical Domains**	**33 / 45**	**2–72**	**2,079,507**	**2,986,574**	**13.72% (avg)**	**Structural, contextual, and point faults**	**33 Primary Held-Out / 12 Case Studies**

TABLE S14

PER-DATASET PRECISION-RECALL AREA UNDER THE CURVE (PR-AUC / AVERAGE PRECISION) ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED CHANNELS). THE BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED. CONTINUOUS RANKING EVALUATES THE UNTHRESHOLDED PREDICTIVE SCORES.

Dataset	**Physical Telemetry Domain**	**Streams**	**TS-JEPA (Control)**	**Operator-Entropy**	**Reynolds-Stress**	**Potential-Flow**	**TimesNet**	**TranAD**
Daphnet	Wearable Parkinsonian Kinematics	6	0.137	0.150	0.151	0.144	**0.343**	0.280
GHL	Gasoil Heating Loop Dynamics	14	**0.016**	0.015	0.011	0.013	0.009	0.010
Genesis	Pick-and-Place Automation	1	**0.053**	0.009	0.005	0.005	0.020	0.006
MSL	NASA Mars Science Laboratory	4	0.427	**0.445**	0.413	0.281	0.310	0.294
SMAP	NASA Soil Moisture Active Passive	6	0.539	**0.591**	0.522	0.261	0.463	0.350
Metro	Interstate Highway Traffic (I-94)	1	0.001	0.001	0.001	0.001	0.002	**0.002**
SWAN	Solar Flare Magnetic Telemetry	1	0.519	0.370	0.397	0.325	0.292	**0.562**
Dataset-Macro Mean	Across All 7 Benchmarks	33	**0.242**	0.227	0.214	0.147	0.206	0.216

TABLE S15

PER-DATASET UNADJUSTED POINT-F1 ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED STREAMS) UNDER HELD-OUT CALIBRATION (SPOT, $q = 10^{-3}$; UP TO FIVE SEEDS ON NON-GHL STREAMS). BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND SECOND BEST IS UNDERLINED. THE BOTTOM ROW REPORTS UNWEIGHTED DATASET-MACRO AVERAGES MATCHING TABLE **??**.

Dataset	**Physical Sensing Domain**	**Streams**	**TS-JEPA (Control)**	**Operator-Entropy**	**Reynolds-Stress**	**Potential-Flow**	**TimesNet**	**TranAD**
Daphnet	Wearable Parkinsonian Kinematics	6	0.125	0.136	0.111	0.057	**0.229**	0.144
GHL	Gasoil Heating Loop Dynamics	14	0.000	0.001	0.001	0.000	**0.022**	0.021
Genesis	Pick-and-Place Automation	1	**0.017**	0.007	0.000	0.002	0.000	0.000
MSL	NASA Mars Science Laboratory	4	0.237	**0.303**	0.219	0.122	0.158	0.151
SMAP	NASA Soil Moisture Active Passive	6	**0.415**	0.401	0.376	0.117	0.289	0.167
Metro	Interstate Highway Traffic (I-94)	1	0.000	0.000	0.001	**0.004**	0.001	0.000
SWAN	Solar Flare Magnetic Telemetry	1	**0.241**	0.060	0.148	0.146	0.000	0.000
Dataset-Macro Mean	Across All 7 Benchmarks	33	**0.148**	0.130	0.122	0.064	0.100	0.069

TABLE S16

NATIVE-WINDOW BASELINE EVALUATION ACROSS FIVE DIVERSE BENCHMARK STREAMS (NASA SMAP: P−3, SMD: MACHINE−1−2, NASA MSL: M−1, DAPHNET: S01R01E1, GECCO: WATER_QUALITY). SHORTENING THE CONTEXT WINDOW TO NATIVE SIZES ($W = 10$ FOR TRANAD, $W = 96$ FOR TIMESNET) EXACERBATES RESIDUAL TAIL HEAVINESS ($\gamma_2 = 448.25$ AND 98.14) RATHER THAN CURING IT.

Model Configuration	**Window W**	**Test γ_2**	**Mean False Pos.**	**Empirical FPR**
TimesNet (Native)	96	98.14	26,526.8	44.09%
TimesNet (Long)	320	32.04	27,041.0	51.26%
TranAD (Native)	10	448.25	18,765.4	24.51%
TranAD (Long)	320	29.93	16,671.0	31.76%
TS-JEPA (Ours)	320	**3.14**	**10,678.8**	**22.57%**

TABLE S17

RESIDUAL-DISTRIBUTION DIAGNOSTICS ON HELD-OUT NOMINAL TELEMETRY (BUFFERED REGIME, STREAM-AVERAGED). γ_2: EXCESS KURTOSIS; $q_{99.9}/q_{99}$: TAIL RATIO; θ: EXTREMAL INDEX (RUNS DECLUSTERING); N_{eff}: EFFECTIVE INDEPENDENT EXCEEDANCES; FPR: OUT-OF-SAMPLE EXCEEDANCE RATE AT THE SPOT THRESHOLD; PR-AUC: ANOMALY-VS-NOMINAL SEPARATION ON LABELED TEST DATA.

Model	γ_2	$q_{99.9}/q_{99}$	ACF_1	θ	N_{eff}	FPR	PR-AUC
JEPA-CtxRecon	1.01	2.300	0.991	0.188	1.7	0.108	0.277
JEPA-NoEMA	1.00	2.187	0.988	0.192	1.8	0.113	0.314
OpEntropy	1.39	2.241	0.989	0.174	1.7	0.083	0.322
PCA-Recon	8.59	1.231	0.998	0.119	1.0	0.312	0.331
PotentialFlow	1.58	2.379	0.985	0.187	1.9	0.093	0.236
Reynolds	1.28	2.039	0.989	0.174	1.8	0.078	0.305
TCN-ObsPred	8.24	1.209	0.998	0.121	1.0	0.310	0.310
TCN-ObsRecon	8.31	1.316	0.998	0.126	1.1	0.303	0.301
TS-JEPA	1.14	2.065	0.991	0.178	1.7	0.080	0.323
TimesNet	8.24	1.309	0.998	0.120	1.0	0.299	0.329
TranAD	10.40	1.220	0.998	0.122	1.1	0.307	0.287
ZScoreMax	9.97	1.617	0.997	0.131	1.2	0.339	0.374

TABLE S18

DETECTION QUALITY AT MATCHED HELD-OUT FALSE-POSITIVE-RATE BUDGETS (STREAM-AVERAGED, q^* = EMPIRICAL CALIBRATION QUANTILE). ER: EVENT RECALL; P: POINT PRECISION; Δt: MEAN DETECTION DELAY; FA/1K: FALSE ALARMS PER 1,000 NOMINAL POINTS; ER_{pre}: RECALL CREDITING PRE-ONSET ALARMS.

Regime (q^*)	Model	ER	P	Δt	FA/1k	ER_{pre}
Buffered 0.001	JEPA-CtxRecon	0.526	0.211	104.5	291.3	0.543
Buffered 0.001	JEPA-NoEMA	0.526	0.273	159.4	308.4	0.552
Buffered 0.001	OpEntropy	0.523	0.264	197.3	247.9	0.529
Buffered 0.001	PCA-Recon	0.584	0.208	45.2	254.4	0.596
Buffered 0.001	PotentialFlow	0.411	0.163	109.5	206.2	0.452
Buffered 0.001	Reynolds	0.442	0.243	139.6	213.4	0.456
Buffered 0.001	TCN-ObsPred	0.582	0.165	213.3	289.8	0.593
Buffered 0.001	TCN-ObsRecon	0.575	0.164	216.2	288.0	0.579
Buffered 0.001	TS-JEPA	0.486	0.277	86.0	207.9	0.510
Buffered 0.001	TimesNet	0.583	0.194	51.5	259.2	0.585
Buffered 0.001	TranAD	0.516	0.130	128.0	262.6	0.554
Buffered 0.001	ZScoreMax	0.609	0.290	77.8	283.2	0.662
Buffered 0.005	JEPA-CtxRecon	0.543	0.195	102.7	303.6	0.571
Buffered 0.005	JEPA-NoEMA	0.552	0.272	148.1	318.1	0.578
Buffered 0.005	OpEntropy	0.536	0.267	191.7	259.2	0.544
Buffered 0.005	PCA-Recon	0.584	0.209	44.8	255.9	0.596
Buffered 0.005	PotentialFlow	0.452	0.174	111.3	218.3	0.487
Buffered 0.005	Reynolds	0.476	0.246	127.0	220.5	0.492
Buffered 0.005	TCN-ObsPred	0.584	0.166	214.6	291.2	0.596
Buffered 0.005	TCN-ObsRecon	0.575	0.164	214.5	289.6	0.583
Buffered 0.005	TS-JEPA	0.503	0.269	84.5	216.3	0.536
Buffered 0.005	TimesNet	0.583	0.194	51.4	260.7	0.585
Buffered 0.005	TranAD	0.516	0.130	127.0	264.6	0.555
Buffered 0.005	ZScoreMax	0.609	0.290	77.6	286.8	0.662
Buffered 0.010	JEPA-CtxRecon	0.560	0.192	98.3	313.5	0.576
Buffered 0.010	JEPA-NoEMA	0.554	0.263	145.0	325.5	0.581
Buffered 0.010	OpEntropy	0.540	0.264	187.6	265.9	0.550
Buffered 0.010	PCA-Recon	0.584	0.210	44.0	258.1	0.598
Buffered 0.010	PotentialFlow	0.468	0.170	102.9	227.4	0.503
Buffered 0.010	Reynolds	0.488	0.243	137.1	227.0	0.504
Buffered 0.010	TCN-ObsPred	0.584	0.167	212.5	293.6	0.598
Buffered 0.010	TCN-ObsRecon	0.575	0.164	211.5	292.1	0.590
Buffered 0.010	TS-JEPA	0.510	0.261	79.2	222.3	0.553
Buffered 0.010	TimesNet	0.590	0.194	50.6	262.7	0.593
Buffered 0.010	TranAD	0.522	0.131	124.4	267.4	0.561
Buffered 0.010	ZScoreMax	0.609	0.290	67.4	295.1	0.662
Buffered 0.050	JEPA-CtxRecon	0.610	0.202	81.6	352.7	0.631
Buffered 0.050	JEPA-NoEMA	0.620	0.243	137.0	362.3	0.645
Buffered 0.050	OpEntropy	0.627	0.248	70.7	314.9	0.640
Buffered 0.050	PCA-Recon	0.597	0.235	51.4	284.5	0.635
Buffered 0.050	PotentialFlow	0.537	0.169	77.0	272.5	0.584
Buffered 0.050	Reynolds	0.543	0.231	111.2	269.3	0.572
Buffered 0.050	TCN-ObsPred	0.595	0.192	205.3	323.1	0.637
Buffered 0.050	TCN-ObsRecon	0.590	0.186	203.0	321.7	0.628
Buffered 0.050	TS-JEPA	0.560	0.246	61.2	258.6	0.612
Buffered 0.050	TimesNet	0.612	0.196	50.0	287.3	0.615
Buffered 0.050	TranAD	0.596	0.145	146.3	295.3	0.622
Buffered 0.050	ZScoreMax	0.609	0.286	63.5	319.5	0.662
Causal 0.001	JEPA-CtxRecon	0.542	0.207	122.1	128.4	–
Causal 0.001	JEPA-NoEMA	0.512	0.256	164.0	126.6	–
Causal 0.001	OpEntropy	0.557	0.232	222.9	103.1	–
Causal 0.001	PCA-Recon	0.589	0.217	206.5	210.2	–
Causal 0.001	PotentialFlow	0.482	0.186	184.6	35.1	–
Causal 0.001	Reynolds	0.486	0.249	159.5	93.0	–
Causal 0.001	TCN-ObsPred	0.546	0.178	46.9	248.7	–
Causal 0.001	TCN-ObsRecon	0.528	0.174	46.9	244.5	–
Causal 0.001	TS-JEPA	0.499	0.261	123.3	70.0	–
Causal 0.001	TimesNet	0.590	0.219	60.1	222.4	–
Causal 0.001	TranAD	0.533	0.156	127.1	209.6	–
Causal 0.001	ZScoreMax	0.587	0.294	62.4	237.9	–
Causal 0.005	JEPA-CtxRecon	0.600	0.211	93.1	149.0	–
Causal 0.005	JEPA-NoEMA	0.599	0.253	124.2	150.1	–
Causal 0.005	OpEntropy	0.634	0.232	81.2	116.7	–
Causal 0.005	PCA-Recon	0.595	0.216	203.9	215.3	–
Causal 0.005	PotentialFlow	0.593	0.194	102.9	46.7	–
Causal 0.005	Reynolds	0.544	0.243	120.8	110.4	–
Causal 0.005	TCN-ObsPred	0.548	0.178	44.8	251.3	–
Causal 0.005	TCN-ObsRecon	0.555	0.174	43.9	248.0	–
Causal 0.005	TS-JEPA	0.628	0.275	104.1	89.2	–
Causal 0.005	TimesNet	0.590	0.219	59.7	226.4	–
Causal 0.005	TranAD	0.536	0.155	126.6	213.0	–
Causal 0.005	ZScoreMax	0.587	0.294	62.4	237.9	–
Causal 0.010	JEPA-CtxRecon	0.640	0.215	75.4	165.7	–
Causal 0.010	JEPA-NoEMA	0.662	0.241	74.9	168.2	–
Causal 0.010	OpEntropy	0.652	0.231	66.8	127.6	–
Causal 0.010	PCA-Recon	0.595	0.219	45.0	221.3	–
Causal 0.010	PotentialFlow	0.637	0.193	70.9	54.8	–
Causal 0.010	Reynolds	0.626	0.250	97.4	125.2	–
Causal 0.010	TCN-ObsPred	0.605	0.180	41.5	255.3	–
Causal 0.010	TCN-ObsRecon	0.595	0.174	40.4	252.4	–
Causal 0.010	TS-JEPA	0.662	0.264	85.7	110.3	–
Causal 0.010	TimesNet	0.595	0.220	63.4	228.8	–

TABLE S19
CONTROLLED-DRIFT CALIBRATION STUDY: ACHIEVED HELD-OUT FPR (DESIGN $q = 10^{-3}$) UNDER INJECTED SHIFTS IN REAL RESIDUAL STREAMS (STREAM-AVERAGED OVER 16 STREAMS WITH ≥ 200 CALIBRATION/EVALUATION WINDOWS; THE THREE SHORT MSL STREAMS ARE EXCLUDED). ROLLING VARIANTS RECALIBRATE PER EVALUATION BLOCK.

Condition	SPOT	SPOT+decl.	p99.5	Roll. p99.5	Roll. SPOT
acf 0.6	0.079	0.182	0.188	0.018	0.051
acf 0.9	0.113	0.218	0.232	0.024	0.055
mean ramp	0.075	0.182	0.186	0.326	0.385
mean shift +0.5s	0.075	0.181	0.186	0.011	0.040
mean shift +1.0s	0.087	0.209	0.216	0.011	0.043
regime change +1s	0.073	0.182	0.187	0.015	0.070
stationary	0.061	0.159	0.162	0.016	0.047
var scale 1.5	0.072	0.172	0.178	0.016	0.047
var scale 2.0	0.082	0.186	0.195	0.016	0.047

TABLE S20
TRAINING-FREE NONPARAMETRIC REFERENCES UNDER THE IDENTICAL HELD-OUT CALIBRATION PROTOCOL (19 NON-GHL STREAMS, FIVE SEEDS, BUFFERED REGIME).

Model	Held-out FPR	Point-F1	PR-AUC	Event Recall	Δt
PCA-Recon	0.312	0.198	0.331	0.584	46.2
ZScoreMax	0.339	0.209	0.374	0.609	67.5

S-9. Complete Statistical Test Map

Table S21 enumerates every paired stream-level comparison reported in the manuscript, resolving Reviewer Concern 5. Two cohorts are reported separately: the 19 non-GHL streams evaluated at five seeds (all paired-inference claims) and the full 33-stream primary benchmark (headline means cited in the abstract). Multiple seeds are averaged within a stream before pairing; the independent statistical unit is the telemetry stream. p_{adj} applies Benjamini–Hochberg FDR control within each cohort $\times$ regime family (45 tests per family). The effect column reports the median paired difference with a 95% bootstrap CI over the stream-level differences; r_b is the rank-biserial correlation.

TABLE S21: Complete statistical test map for paired stream-level comparisons (Wilcoxon signed-rank on stream-macro means, seeds averaged within stream).

Cohort	Regime	Endpoint	Comparison	n	p	p_{adj}	Median diff. [CI]	r_b
non-GHL (n=19)	Buffered	Held-out nominal FPR	TS-JEPA - TimesNet	19	0.0401	0.2478	-0.0713 [-0.3759, -0.0833]	-0.16
		Held-out nominal FPR	TS-JEPA - TranAD	19	0.0258	0.2478	-0.1511 [-0.3714, -0.0942]	-0.16
		Held-out nominal FPR	OpEntropy - TS-JEPA	19	0.9843	0.9843	-0.0050 [-0.0278, 0.0485]	-0.16
		Held-out nominal FPR	Reynolds - TS-JEPA	19	0.9217	0.9427	0.0004 [-0.0380, 0.0453]	0.05
		Held-out nominal FPR	PotentialFlow - TS-JEPA	19	0.7381	0.9427	0.0017 [-0.0259, 0.0556]	0.05
		Held-out nominal FPR	OpEntropy - TimesNet	19	0.0352	0.2478	-0.0566 [-0.3646, -0.0843]	-0.18
		Held-out nominal FPR	Reynolds - TimesNet	19	0.0198	0.2478	-0.0576 [-0.3536, -0.0939]	-0.22
		Held-out nominal FPR	PotentialFlow - TimesNet	19	0.0494	0.2478	-0.0857 [-0.3577, -0.0625]	-0.16
		Held-out nominal FPR	TimesNet - TranAD	19	0.8139	0.9427	0.0000 [-0.1179, 0.0986]	0.00
		Point-F1	TS-JEPA - TimesNet	19	0.5678	0.9162	0.0168 [-0.0702, 0.1451]	0.05
		Point-F1	TS-JEPA - TranAD	19	0.2145	0.5081	0.0168 [-0.0094, 0.2289]	0.11
		Point-F1	OpEntropy - TS-JEPA	19	0.6791	0.9427	-0.0093 [-0.0428, 0.0512]	-0.22
		Point-F1	Reynolds - TS-JEPA	19	0.4413	0.7944	-0.0019 [-0.0754, 0.0125]	-0.05
		Point-F1	PotentialFlow - TS-JEPA	19	0.0108	0.2478	-0.0946 [-0.2712, -0.0478]	-0.47
		Point-F1	OpEntropy - TimesNet	19	0.2579	0.5527	0.0283 [-0.0679, 0.1416]	0.26
		Point-F1	Reynolds - TimesNet	19	0.8789	0.9427	0.0000 [-0.0971, 0.1160]	0.00
		Point-F1	PotentialFlow - TimesNet	19	0.1447	0.4069	-0.0295 [-0.2327, -0.0070]	-0.16
		Point-F1	TimesNet - TranAD	19	0.0640	0.2478	0.0000 [0.0180, 0.1303]	0.29
		PR-AUC	TS-JEPA - TimesNet	19	0.9217	0.9427	-0.0023 [-0.1733, 0.1570]	-0.16
		PR-AUC	TS-JEPA - TranAD	19	0.6794	0.9427	0.0015 [-0.1254, 0.1996]	0.05
		PR-AUC	OpEntropy - TS-JEPA	19	0.8288	0.9427	-0.0040 [-0.0357, 0.0362]	-0.26
		PR-AUC	Reynolds - TS-JEPA	19	0.1232	0.3696	-0.0177 [-0.0445, 0.0152]	-0.26
		PR-AUC	PotentialFlow - TS-JEPA	19	0.1134	0.3644	-0.0090 [-0.2182, -0.0297]	-0.26
		PR-AUC	OpEntropy - TimesNet	19	0.9217	0.9427	-0.0003 [-0.1748, 0.1464]	-0.05
		PR-AUC	Reynolds - TimesNet	19	0.8288	0.9427	-0.0142 [-0.1845, 0.1352]	-0.16
		PR-AUC	PotentialFlow - TimesNet	19	0.2101	0.5081	-0.0134 [-0.2620, -0.0101]	-0.16
		PR-AUC	TimesNet - TranAD	19	0.0361	0.2478	0.0116 [-0.0207, 0.1166]	0.37
		Event recall	TS-JEPA - TimesNet	19	0.3938	0.7383	-0.0080 [-0.3681, 0.1526]	-0.18
		Event recall	TS-JEPA - TranAD	19	0.9056	0.9427	0.0000 [-0.2633, 0.2440]	0.06
		Event recall	OpEntropy - TS-JEPA	19	0.8017	0.9427	0.0000 [-0.0525, 0.1148]	-0.14
		Event recall	Reynolds - TS-JEPA	19	0.3937	0.7383	0.0000 [-0.1710, 0.0905]	-0.06
		Event recall	PotentialFlow - TS-JEPA	19	0.0533	0.2478	0.0000 [-0.3347, -0.0341]	-0.20
		Event recall	OpEntropy - TimesNet	19	0.6050	0.9388	0.0000 [-0.3026, 0.1440]	0.00
		Event recall	Reynolds - TimesNet	19	0.1863	0.4931	-0.0667 [-0.3547, 0.0734]	-0.25
		Event recall	PotentialFlow - TimesNet	19	0.0610	0.2478	-0.4000 [-0.5145, -0.0525]	-0.16
		Event recall	TimesNet - TranAD	19	0.0661	0.2478	0.0000 [0.0022, 0.2007]	0.56
		Detection delay (steps)	TS-JEPA - TimesNet	11	0.2783	0.5693	28.4000 [-36.5661, 67.4297]	0.45
		Detection delay (steps)	TS-JEPA - TranAD	10	0.6953	0.9427	32.1970 [-626.3884, 52.6033]	0.40
		Detection delay (steps)	OpEntropy - TS-JEPA	16	0.5701	0.9162	4.5180 [-21.4413, 347.5172]	0.20
		Detection delay (steps)	Reynolds - TS-JEPA	14	0.6698	0.9427	3.9517 [-15.1168, 105.9640]	0.00
		Detection delay (steps)	PotentialFlow - TS-JEPA	15	0.8753	0.9427	-2.0000 [-22.4649, 38.7592]	-0.14
		Detection delay (steps)	OpEntropy - TimesNet	13	0.0942	0.3262	39.9176 [-3.4980, 116.3097]	0.38
		Detection delay (steps)	Reynolds - TimesNet	13	0.0210	0.2478	67.0000 [33.0242, 183.0271]	0.50
		Detection delay (steps)	PotentialFlow - TimesNet	13	0.2334	0.5251	43.8857 [-19.9464, 330.3680]	0.50
		Detection delay (steps)	TimesNet - TranAD	12	0.4961	0.8586	0.0000 [-534.4812, 9.8618]	-0.11
non-GHL (n=19)	Causal	Held-out nominal FPR	TS-JEPA - TimesNet	18	0.0086	0.0669	-0.1426 [-0.3580, -0.1185]	-0.29
		Held-out nominal FPR	TS-JEPA - TranAD	18	0.0086	0.0669	-0.1685 [-0.3271, -0.1053]	-0.29
		Held-out nominal FPR	OpEntropy - TS-JEPA	18	0.7960	0.8330	0.0004 [-0.0044, 0.0210]	0.12
		Held-out nominal FPR	Reynolds - TS-JEPA	18	0.5862	0.7834	-0.0002 [-0.0103, 0.0039]	-0.06
		Held-out nominal FPR	PotentialFlow - TS-JEPA	18	0.3692	0.5729	-0.0006 [-0.0140, 0.0036]	-0.33
		Held-out nominal FPR	OpEntropy - TimesNet	18	0.0086	0.0669	-0.1386 [-0.3531, -0.1135]	-0.29
		Held-out nominal FPR	Reynolds - TimesNet	18	0.0104	0.0669	-0.1263 [-0.3537, -0.1158]	-0.22
		Held-out nominal FPR	PotentialFlow - TimesNet	18	0.0104	0.0669	-0.1409 [-0.3593, -0.1155]	-0.22
		Held-out nominal FPR	TimesNet - TranAD	18	0.7671	0.8219	0.0000 [-0.0341, 0.1041]	-0.11
		Point-F1	TS-JEPA - TimesNet	18	0.1187	0.2420	-0.0336 [-0.1980, -0.0031]	-0.22
		Point-F1	TS-JEPA - TranAD	18	0.6397	0.7834	-0.0129 [-0.1352, 0.0725]	-0.11
		Point-F1	OpEntropy - TS-JEPA	18	0.5226	0.7586	-0.0027 [-0.0316, 0.0277]	-0.11
		Point-F1	Reynolds - TS-JEPA	18	0.0898	0.2376	-0.0081 [-0.0609, 0.0097]	-0.44
		Point-F1	PotentialFlow - TS-JEPA	18	0.0090	0.0669	-0.0146 [-0.1523, -0.0349]	-0.44
		Point-F1	OpEntropy - TimesNet	18	0.0840	0.2363	-0.0252 [-0.1886, -0.0111]	-0.29
		Point-F1	Reynolds - TimesNet	18	0.0245	0.1104	-0.0842 [-0.2134, -0.0354]	-0.41
		Point-F1	PotentialFlow - TimesNet	18	0.0015	0.0661	-0.0801 [-0.3002, -0.0854]	-0.73
		Point-F1	TimesNet - TranAD	18	0.0392	0.1472	0.0008 [0.0136, 0.1444]	0.38
		PR-AUC	TS-JEPA - TimesNet	18	0.7660	0.8219	-0.0052 [-0.1437, 0.0892]	-0.11
		PR-AUC	TS-JEPA - TranAD	18	0.7019	0.7896	0.0093 [-0.0953, 0.1395]	0.11
		PR-AUC	OpEntropy - TS-JEPA	18	0.0599	0.1927	-0.0050 [-0.0436, 0.0167]	-0.56
		PR-AUC	Reynolds - TS-JEPA	18	0.1964	0.3535	-0.0054 [-0.0295, 0.0425]	-0.33

continued

(continued)

Cohort	Regime	Endpoint	Comparison	n	p	p_{adj}	Median diff. [CI]	r_b
		PR-AUC	PotentialFlow - TS-JEPA	18	0.4171	0.6257	-0.0010 [-0.1788, -0.0127]	0.00
		PR-AUC	OpEntropy - TimesNet	18	0.6397	0.7834	-0.0010 [-0.1547, 0.0666]	-0.11
		PR-AUC	Reynolds - TimesNet	18	0.6095	0.7834	-0.0046 [-0.1471, 0.0947]	-0.22
		PR-AUC	PotentialFlow - TimesNet	18	0.2462	0.4261	-0.0035 [-0.2195, -0.0287]	0.00
		PR-AUC	TimesNet - TranAD	18	0.0304	0.1242	0.0086 [-0.0201, 0.1242]	0.44
		Event recall	TS-JEPA - TimesNet	18	0.2656	0.4427	-0.1000 [-0.2926, 0.0496]	-0.06
		Event recall	TS-JEPA - TranAD	18	0.5418	0.7619	-0.0889 [-0.2420, 0.1024]	0.00
		Event recall	OpEntropy - TS-JEPA	18	0.9176	0.9384	-0.0040 [-0.0920, 0.0954]	-0.12
		Event recall	Reynolds - TS-JEPA	18	0.9645	0.9645	0.0000 [-0.1117, 0.0853]	-0.09
		Event recall	PotentialFlow - TS-JEPA	18	0.1237	0.2420	-0.1000 [-0.2694, 0.0323]	-0.53
		Event recall	OpEntropy - TimesNet	18	0.2802	0.4504	0.0000 [-0.3077, 0.0661]	-0.07
		Event recall	Reynolds - TimesNet	18	0.1053	0.2420	-0.2000 [-0.2770, 0.0233]	-0.33
		Event recall	PotentialFlow - TimesNet	18	0.0156	0.0781	-0.0962 [-0.4292, -0.0750]	-0.43
		Event recall	TimesNet - TranAD	18	0.1159	0.2420	0.0000 [-0.0125, 0.1420]	0.67
		Detection delay (steps)	TS-JEPA - TimesNet	13	0.0681	0.2043	45.6000 [-2.1414, 137.9843]	0.54
		Detection delay (steps)	TS-JEPA - TranAD	13	0.1465	0.2747	58.8000 [-468.1920, 132.5104]	0.69
		Detection delay (steps)	OpEntropy - TS-JEPA	17	0.6441	0.7834	1.1667 [-38.0739, 25.5760]	0.06
		Detection delay (steps)	Reynolds - TS-JEPA	17	0.0980	0.2420	16.4667 [7.1296, 103.0613]	0.38
		Detection delay (steps)	PotentialFlow - TS-JEPA	14	0.6698	0.7896	-7.6306 [-51.3326, 48.5405]	-0.14
		Detection delay (steps)	OpEntropy - TimesNet	13	0.0479	0.1656	75.0000 [-4.1865, 117.4269]	0.54
		Detection delay (steps)	Reynolds - TimesNet	13	0.0134	0.0755	39.9631 [36.8358, 234.6891]	0.69
		Detection delay (steps)	PotentialFlow - TimesNet	12	0.1099	0.2420	68.7500 [0.8355, 113.9374]	0.50
		Detection delay (steps)	TimesNet - TranAD	13	0.6953	0.7896	0.0000 [-495.3941, 16.5770]	-0.20
primary (n=33)	Buffered	Held-out nominal FPR	TS-JEPA - TimesNet	33	0.0024	0.0179	-0.0378 [-0.2454, -0.0665]	-0.33
		Held-out nominal FPR	TS-JEPA - TranAD	33	0.0029	0.0187	-0.0636 [-0.2439, -0.0670]	-0.33
		Held-out nominal FPR	OpEntropy - TS-JEPA	33	0.7509	0.8604	-0.0007 [-0.0131, 0.0316]	-0.03
		Held-out nominal FPR	Reynolds - TS-JEPA	33	0.9158	0.9158	-0.0008 [-0.0229, 0.0249]	-0.03
		Held-out nominal FPR	PotentialFlow - TS-JEPA	33	0.3215	0.5358	0.0027 [-0.0130, 0.0341]	0.27
		Held-out nominal FPR	OpEntropy - TimesNet	33	0.0061	0.0304	-0.0514 [-0.2404, -0.0621]	-0.29
		Held-out nominal FPR	Reynolds - TimesNet	33	0.0011	0.0103	-0.0534 [-0.2417, -0.0742]	-0.44
		Held-out nominal FPR	PotentialFlow - TimesNet	33	0.0078	0.0352	-0.0562 [-0.2361, -0.0539]	-0.27
		Held-out nominal FPR	TimesNet - TranAD	33	0.2641	0.4755	0.0000 [-0.0654, 0.0671]	0.28
		Point-F1	TS-JEPA - TimesNet	33	0.5372	0.8058	-0.0044 [-0.0497, 0.0764]	-0.38
		Point-F1	TS-JEPA - TranAD	33	0.7648	0.8604	-0.0050 [-0.0138, 0.1286]	-0.38
		Point-F1	OpEntropy - TS-JEPA	33	0.7089	0.8577	0.0000 [-0.0245, 0.0305]	-0.10
		Point-F1	Reynolds - TS-JEPA	33	0.6143	0.8134	0.0000 [-0.0431, 0.0079]	0.05
		Point-F1	PotentialFlow - TS-JEPA	33	0.0325	0.1180	0.0000 [-0.1581, -0.0213]	-0.33
		Point-F1	OpEntropy - TimesNet	33	0.9107	0.9158	-0.0041 [-0.0469, 0.0731]	-0.25
		Point-F1	Reynolds - TimesNet	33	0.2557	0.4755	-0.0045 [-0.0656, 0.0588]	-0.35
		Point-F1	PotentialFlow - TimesNet	33	0.0387	0.1245	-0.0097 [-0.1456, -0.0087]	-0.39
		Point-F1	TimesNet - TranAD	33	0.1271	0.3443	0.0000 [0.0096, 0.0785]	0.07
		PR-AUC	TS-JEPA - TimesNet	33	0.4473	0.7034	0.0024 [-0.0968, 0.0887]	0.21
		PR-AUC	TS-JEPA - TranAD	33	0.2075	0.4284	0.0024 [-0.0723, 0.1187]	0.33
		PR-AUC	OpEntropy - TS-JEPA	33	0.8601	0.9158	-0.0008 [-0.0190, 0.0219]	-0.15
		PR-AUC	Reynolds - TS-JEPA	33	0.1301	0.3443	-0.0033 [-0.0255, 0.0116]	-0.27
		PR-AUC	PotentialFlow - TS-JEPA	33	0.5843	0.8134	0.0002 [-0.1304, -0.0129]	0.03
		PR-AUC	OpEntropy - TimesNet	33	0.2565	0.4755	0.0035 [-0.0944, 0.0876]	0.21
		PR-AUC	Reynolds - TimesNet	33	0.7243	0.8577	0.0011 [-0.1064, 0.0801]	0.03
		PR-AUC	PotentialFlow - TimesNet	33	0.8601	0.9158	0.0022 [-0.1545, 0.0004]	0.21
		PR-AUC	TimesNet - TranAD	33	0.1495	0.3542	0.0003 [-0.0138, 0.0690]	0.03
		Event recall	TS-JEPA - TimesNet	33	0.0003	0.0044	-0.8000 [-0.6479, -0.2364]	-0.53
		Event recall	TS-JEPA - TranAD	33	0.0034	0.0190	-0.5000 [-0.5705, -0.1469]	-0.42
		Event recall	OpEntropy - TS-JEPA	33	0.4533	0.7034	0.0000 [-0.0224, 0.0816]	0.00
		Event recall	Reynolds - TS-JEPA	33	0.6146	0.8134	0.0000 [-0.0930, 0.0655]	0.05
		Event recall	PotentialFlow - TS-JEPA	33	0.1623	0.3653	0.0000 [-0.1887, 0.0049]	-0.06
		Event recall	OpEntropy - TimesNet	33	0.0006	0.0070	-0.6000 [-0.6084, -0.2193]	-0.45
		Event recall	Reynolds - TimesNet	33	0.0003	0.0044	-0.6000 [-0.6430, -0.2752]	-0.59
		Event recall	PotentialFlow - TimesNet	33	$<$0.0001	0.0032	-0.8000 [-0.7058, -0.3426]	-0.45
		Event recall	TimesNet - TranAD	33	0.0341	0.1180	0.0000 [0.0159, 0.1528]	0.50
		Detection delay (steps)	TS-JEPA - TimesNet	11	0.2783	0.4817	28.4000 [-38.1938, 65.4996]	0.45
		Detection delay (steps)	TS-JEPA - TranAD	10	0.6953	0.8577	32.1970 [-627.2807, 54.1011]	0.40
		Detection delay (steps)	OpEntropy - TS-JEPA	16	0.5701	0.8134	4.5180 [-22.0503, 349.4281]	0.20
		Detection delay (steps)	Reynolds - TS-JEPA	14	0.6698	0.8577	3.9517 [-14.5613, 110.2057]	0.00
		Detection delay (steps)	PotentialFlow - TS-JEPA	15	0.8753	0.9158	-2.0000 [-22.8252, 37.3496]	-0.14
		Detection delay (steps)	OpEntropy - TimesNet	15	0.0480	0.1440	39.9176 [6.2208, 121.1460]	0.43
		Detection delay (steps)	Reynolds - TimesNet	15	0.0303	0.1180	67.0000 [33.7361, 254.5029]	0.43
		Detection delay (steps)	PotentialFlow - TimesNet	14	0.2094	0.4284	35.0845 [-17.2960, 313.0068]	0.50
		Detection delay (steps)	TimesNet - TranAD	25	0.1477	0.3542	0.0000 [-304.7382, 13.9861]	-0.25
primary (n=33)	Causal	Held-out nominal FPR	TS-JEPA - TimesNet	32	0.0002	0.0013	-0.0549 [-0.2354, -0.0804]	-0.42
		Held-out nominal FPR	TS-JEPA - TranAD	32	0.0002	0.0013	-0.0517 [-0.2094, -0.0707]	-0.42
		Held-out nominal FPR	OpEntropy - TS-JEPA	32	0.3820	0.5372	0.0008 [-0.0015, 0.0151]	0.27
		Held-out nominal FPR	Reynolds - TS-JEPA	32	0.5052	0.6315	-0.0004 [-0.0059, 0.0018]	-0.10
		Held-out nominal FPR	PotentialFlow - TS-JEPA	32	0.7190	0.7670	-0.0004 [-0.0074, 0.0029]	-0.12
		Held-out nominal FPR	OpEntropy - TimesNet	32	0.0004	0.0019	-0.0550 [-0.2243, -0.0791]	-0.35
		Held-out nominal FPR	Reynolds - TimesNet	32	$<$0.0001	0.0013	-0.0558 [-0.2360, -0.0821]	-0.44
		Held-out nominal FPR	PotentialFlow - TimesNet	32	0.0002	0.0013	-0.0556 [-0.2374, -0.0842]	-0.31
		Held-out nominal FPR	TimesNet - TranAD	32	0.4264	0.5717	0.0000 [-0.0132, 0.0664]	0.27
		Point-F1	TS-JEPA - TimesNet	32	0.0135	0.0435	-0.0068 [-0.1229, -0.0087]	-0.55
		Point-F1	TS-JEPA - TranAD	32	0.0811	0.1826	-0.0065 [-0.0824, 0.0352]	-0.48
		Point-F1	OpEntropy - TS-JEPA	32	0.6580	0.7351	0.0000 [-0.0183, 0.0165]	-0.05
		Point-F1	Reynolds - TS-JEPA	32	0.1590	0.2523	0.0000 [-0.0333, 0.0064]	-0.37

continued

(continued)

Cohort	Regime	Endpoint	Comparison	n	p	p_{adj}	Median diff. [CI]	r_b
		Point-F1	PotentialFlow - TS-JEPA	32	0.0400	0.1126	0.0000 [-0.0876, -0.0140]	-0.30
		Point-F1	OpEntropy - TimesNet	32	0.0093	0.0321	-0.0074 [-0.1223, -0.0153]	-0.60
		Point-F1	Reynolds - TimesNet	32	0.0032	0.0119	-0.0187 [-0.1342, -0.0265]	-0.60
		Point-F1	PotentialFlow - TimesNet	32	0.0002	0.0013	-0.0245 [-0.1874, -0.0493]	-0.72
		Point-F1	TimesNet - TranAD	32	0.0775	0.1826	0.0000 [0.0091, 0.0885]	0.00
		PR-AUC	TS-JEPA - TimesNet	32	0.4883	0.6278	0.0020 [-0.0792, 0.0536]	0.25
		PR-AUC	TS-JEPA - TranAD	32	0.1195	0.2150	0.0040 [-0.0508, 0.0793]	0.38
		PR-AUC	OpEntropy - TS-JEPA	32	0.1545	0.2523	-0.0013 [-0.0240, 0.0111]	-0.31
		PR-AUC	Reynolds - TS-JEPA	32	0.2699	0.4021	-0.0030 [-0.0151, 0.0255]	-0.31
		PR-AUC	PotentialFlow - TS-JEPA	32	0.8034	0.8216	0.0014 [-0.1044, -0.0039]	0.19
		PR-AUC	OpEntropy - TimesNet	32	0.4319	0.5717	0.0022 [-0.0838, 0.0417]	0.19
		PR-AUC	Reynolds - TimesNet	32	0.7329	0.7670	0.0002 [-0.0800, 0.0580]	0.00
		PR-AUC	PotentialFlow - TimesNet	32	0.6247	0.7351	0.0031 [-0.1216, -0.0081]	0.31
		PR-AUC	TimesNet - TranAD	32	0.1195	0.2150	0.0003 [-0.0116, 0.0681]	0.06
		Event recall	TS-JEPA - TimesNet	32	0.0003	0.0013	-0.4462 [-0.6130, -0.2721]	-0.47
		Event recall	TS-JEPA - TranAD	32	0.0007	0.0029	-0.3667 [-0.5717, -0.2184]	-0.42
		Event recall	OpEntropy - TS-JEPA	32	0.6528	0.7351	0.0000 [-0.0436, 0.0676]	-0.06
		Event recall	Reynolds - TS-JEPA	32	0.8444	0.8444	0.0000 [-0.0586, 0.0554]	0.00
		Event recall	PotentialFlow - TS-JEPA	32	0.2770	0.4021	0.0000 [-0.1508, 0.0344]	-0.37
		Event recall	OpEntropy - TimesNet	32	0.0003	0.0013	-0.5167 [-0.6028, -0.2453]	-0.50
		Event recall	Reynolds - TimesNet	32	0.0001	0.0013	-0.4462 [-0.6063, -0.2734]	-0.64
		Event recall	PotentialFlow - TimesNet	32	$<$0.0001	0.0013	-0.5778 [-0.6447, -0.3384]	-0.64
		Event recall	TimesNet - TranAD	32	0.1095	0.2149	0.0000 [-0.0082, 0.1126]	0.64
		Detection delay (steps)	TS-JEPA - TimesNet	13	0.0681	0.1703	45.6000 [-0.1247, 136.8621]	0.54
		Detection delay (steps)	TS-JEPA - TranAD	13	0.1465	0.2523	58.8000 [-469.1299, 126.9832]	0.69
		Detection delay (steps)	OpEntropy - TS-JEPA	17	0.6441	0.7351	1.1667 [-37.9789, 25.0056]	0.06
		Detection delay (steps)	Reynolds - TS-JEPA	17	0.0980	0.2100	16.4667 [6.3093, 103.6322]	0.38
		Detection delay (steps)	PotentialFlow - TS-JEPA	14	0.6698	0.7351	-7.6306 [-49.2915, 48.7951]	-0.14
		Detection delay (steps)	OpEntropy - TimesNet	14	0.0464	0.1228	57.9500 [-1.1657, 109.9677]	0.54
		Detection delay (steps)	Reynolds - TimesNet	14	0.0353	0.1058	38.7815 [23.5277, 215.7128]	0.57
		Detection delay (steps)	PotentialFlow - TimesNet	13	0.1099	0.2149	62.5000 [-0.4251, 105.6324]	0.50
		Detection delay (steps)	TimesNet - TranAD	26	0.1626	0.2523	0.0000 [-300.2782, 3.3327]	-0.29

REFERENCES

[1] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[2] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[5] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[6] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[7] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.